



GyaanVault

SPPU M.Sc. Data Science — Free Study Material

Deep Learning

DS-562-MJ

Sample / Practice Question Paper

Semester II · Elective · Practice Paper

Time: 3 Hours

Maximum Marks: 35

Instructions: (1) All questions carry equal weightage where indicated. (2) Figures to the right indicate full marks. (3) Draw neat diagrams / write code wherever necessary.

Q1. Answer any FIVE of the following. [5 × 2 = 10]

- a) What is a perceptron?
- b) What is backpropagation?
- c) Name three activation functions.
- d) What is a CNN?
- e) Define convolution and pooling.
- f) What is transfer learning?
- g) What is an LSTM?

Q2. Answer the following. [10]

A) Explain the perceptron, multilayer perceptron and backpropagation.

OR

B) Explain gradient descent, stochastic gradient descent and optimizers.

Q3. Answer the following. [10]

A) Explain CNN architecture: convolution, pooling, strides and transfer learning.

OR

B) Explain RNNs, LSTM and GRU with applications.

Q4. Answer the following. [5]

A) Explain reinforcement learning, MDP and Q-learning.

OR

B) Explain ethical considerations, bias and fairness in deep learning.

